

Experimental Methods to study astronomical objects

Measuring Cosmic Expansion: Galaxy Redshifts with a High-Resolution Spectroscope

Spectroscopic Analysis of galaxies M82 and M31

Research Plan

Spectroscopic analysis of celestial objects is a powerful tool in astrophysics, allowing us to investigate the physical properties, composition, and behavior of observable objects in space. Methods of spectroscopy reveals a wealth of information that is otherwise invisible to the naked eye or by merely observing the brightness or position in the sky.

One such category of celestial objects, the galaxies, are very common in the universe. There are so many in the observable universe, that it is impossible to be able to catalog and study them. Also, the capability to analyze chemical, physical and biological properties are so limited. What we plan to undertake as a part of our plan is to be able to analyze two distinct galaxies, M82 and M31. These are two distinct galaxies presenting totally contrasting physical and chemical properties. As a part of this research, we plan to be able to analyze spectroscopic data, independently, of these two galaxies, and draw conclusions on the following topics:

1. **Radial Velocity Estimation:** By calculating the shift in spectra, we should be able to determine if the radial velocity with respect to the heliocentric position. We should be able to determine if the galaxy is speeding away from our relative position or speeding towards us.
2. **Orbital Parameters:** Spectroscopic observations can reveal the radial velocity of each component as a function of time. This data, when combined with Kepler's laws, allows for the determination of orbital parameters such as the orbital period, semi-major axis, eccentricity, and inclination.

Research Question

The highlighted research questions focus on gaining deeper insights into the fundamental physical chemical processes of the two identified galaxies (M31 and M82) Based on all research materials, it is estimated one is blue-shifted and the other redshifted, one has total absence of absorption spectra and the other has abundance of the same. One is a cradle of very hot and bright, short lived stars, while the other is not. Spectroscopy is a key tool in answering these questions, as it provides direct measurements of velocity, chemical composition, temperature, and other vital parameters to be able to draw conclusions.

While research involving the spectroscopic analysis of galaxies can address a wide range of questions related to stellar properties, dynamics, and evolution, we are limited by the state of equipments, terrestrial locations of observations and most importantly the quality of data. Hence, our research question would be primarily focused on the following topics:

1. Galactic Atmosphere and Chemical Composition

- What are the chemical abundances of the two studied galaxies. How do they compare from one to the other
- Are there differences in the galactic atmospheres of the two components due to the presence of enormous HII elements, presence of forbidden H, N or S

2. Determining the spectral shift accurately and deducing radial velocity

- Experimental and analytical methods to accurately determine stellar shifts of wavelengths of the two galactic systems
- Determining the radial velocity in relation to the position of Earth

3. Determining distances of the the two galaxies

- Is it possible to determine the distances of these galaxies from our relative position
- accurately determine if the galaxy is heading towards us or away from us

Hypothesis

The hypothesis is to be able to perform Spectroscopic analysis of M82 and M31 which will:

- Reveal clear signatures of acceleration of M82, away from the relative position of earth
- Reveal clear signatures of acceleration of M31, towards the relative position of earth
- The distance of M82 is many million light years while that of M31 is just a couple
- M82 is a starburst galaxy with mass ejections of ionized HII and NII gasses, while M31 is a relatively stable galaxy
- M82 is a cradle of young stars proving the existence of HII Region, while M31 does not have such indications

Procedure

The **Shelyak UVEX** spectrograph, is a high-resolution, fiber-fed spectrograph designed for use in astronomical research, particularly for the analysis of stars,

including binary systems. If you're conducting research on binary stars using the UVEX spectrograph, the general procedure would involve several key steps: observation planning, data collection, data reduction, and analysis. Below is an outline of the procedure:

Instrumentation used:

- Telescope(s) Takahashi apochromatic refractor, 106 mm aperture and 530 mm focal length; Meade Schmidt Cassegrain 305 mm aperture and 3,048 mm focal length
- Mount: Multiple computerized German Equatorial Mounts
- Home made, 3d printed, Czerny-Turner spectrograph optimized for wavelengths from near UV (<350nm) to near IR (1 μ m) with multiple diffraction gratings
- Various CCD cameras, photometry and spectroscopy software, data analysis software

1. Objective Definition and Planning

- **Research Question:** For research questions described above: orbital dynamics of a binary system, determining stellar compositions, analyzing mass transfer in a close binary system
 - **Target Selection:** The binary star system(s) to observe based on the scientific goal is yet to be determined. The following characteristics, such as its orbital period, spectral type, distance, and brightness for known eclipsing binaries or spectroscopic binaries with known radial velocity curves will be ideal candidates.
 - **Observation Strategy:** An overall plan for observations will include:
 - Deciding on the wavelength range to observe, based on the expected spectral features (range from 300 to 600 nm).
 - Choosing the appropriate exposure times and observing cadence, especially for systems that show periodic variations (e.g., eclipses or orbital motion).

2. Data Collection

- **Setup Instrumentation (OTA, Spectrograph, mounts, cameras, etc.):**
 - Install the spectrograph onto the telescope, ensuring proper alignment and calibration.
 - **Calibration:**
 - Before collecting data, the optical train will be calibrated to ensure accurate wavelength measurements. This may include:
 - **Flat-field calibration:** Ensuring the uniformity of the spectral response.
 - **Wavelength calibration:** Using a known light source (e.g., a argon-neon calibration lamp with known emission lines) to map pixel positions to specific wavelengths.
- **Observe the Binary System**

- Pointing the telescope at the binary star system using a computerized GEM and beginning the data collection process.
- Ensuring that the spectral resolution is adequate to detect the features needed (e.g., Doppler shifts in spectral lines for radial velocity measurements).
- Multiple observations over time will be needed to capture variations such as eclipses, radial velocity changes, or light curves.

3. Data Reduction

- **Raw Data Processing:** Once data is collected, the raw spectra will be processed to correct for instrumental effects.
- **Bias and Dark Current Subtraction:** Removing any baseline noise or offsets.
- **Flat-field Correction:** Correction for any pixel-to-pixel sensitivity variations across the detector.
- **Wavelength Calibration:** Using the reference lamp spectra, applying a wavelength solution to the observed spectra.
- **Extraction of Spectral Data:**
 - Extracting the one-dimensional spectrum from the two-dimensional image (the spectral data will be recorded as a series of pixels).
 - Applying any necessary sky subtraction for background contamination in the home observatory
- **Correction for Atmospheric Effects:** If necessary, apply any corrections for telluric absorption (especially in the UV-visible range) to ensure that we are detecting stellar features rather than atmospheric lines.
- **Correction for Instrument response:** If necessary, apply any corrections for data losses due to light path within the OTA.
- **Normalize the Spectrum:** Normalization of the spectrum to a consistent intensity to make it easier to analyze spectral features and compare across different observations.

4. Data Analysis

- **Radial Velocity Measurement:** For both the galaxies, the primary analysis will typically focused on radial velocity measurements. This will be done by identifying Doppler shifts in the absorption lines of the spectra.
 - Measuring the displacement of spectral lines from their rest positions (using features like hydrogen Balmer lines or metallic absorption lines).
 - Computing the radial velocities of both galaxies from the shifts in their respective lines.
 - Tracking how these velocities change over time to derive the orbital parameters (e.g., orbital period, eccentricity, semi-major axis).
- **Spectral Features and Stellar Properties:** Analyzing spectral features to determine the chemical properties of the two galaxy
 - Measurement of the spectral type and luminosity class of each galaxy by comparing observed features (e.g., absorption lines, broadening of lines due to

rotation) to standard templates.

- Comparing observed chemical abundances (e.g., metallicity and forbidden elements) for each galaxy
- **Mass Transfer and Accretion:** Detection of any high-velocity emission lines (e.g., H-alpha) or broadening of lines indicative of accretion or outflows.

5. Interpretation and Conclusion

- **Model Fitting:** Use the collected radial velocity data and spectral analysis to fit models of the two observed galaxies based on research hypothesis.
- **Comparison with Theoretical Models:** Comparing the results with theoretical models of spectral shifts, radial velocities, absorption patterns and star formations. This will validate our findings or highlight discrepancies that need further investigation.
- **Reporting Results:** Compiling our findings into a comprehensive analysis, including orbital parameters, stellar properties, and any notable phenomena (such as eclipses, accretion, or unusual spectral features).

7. Conclusion

- Once our data has been reduced and analyzed, the results will be used to draw conclusions on the two observed galaxies of M31 and M82. Highlight If the system exhibits interesting features (e.g., unusual spectral lines, unexpected radial velocities, abundance of forbidden elements, absence of absorption lines), these can provide valuable insights into the physical processes at play in these galactic systems

Data Analysis

As highlighted in the procedure section, following data analysis technique will be used:

Radial Velocity Measurement: For both the galaxies, the primary analysis will typically focused on radial velocity measurements. This will be done by identifying Doppler shifts in the absorption lines of the spectra.

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